Module 9, Lesson 2

AI 101 FOR MECHANICAL ENGINEERS

Trees and Ensembles: Undergraduate Course

Gemini Notebook

Outline

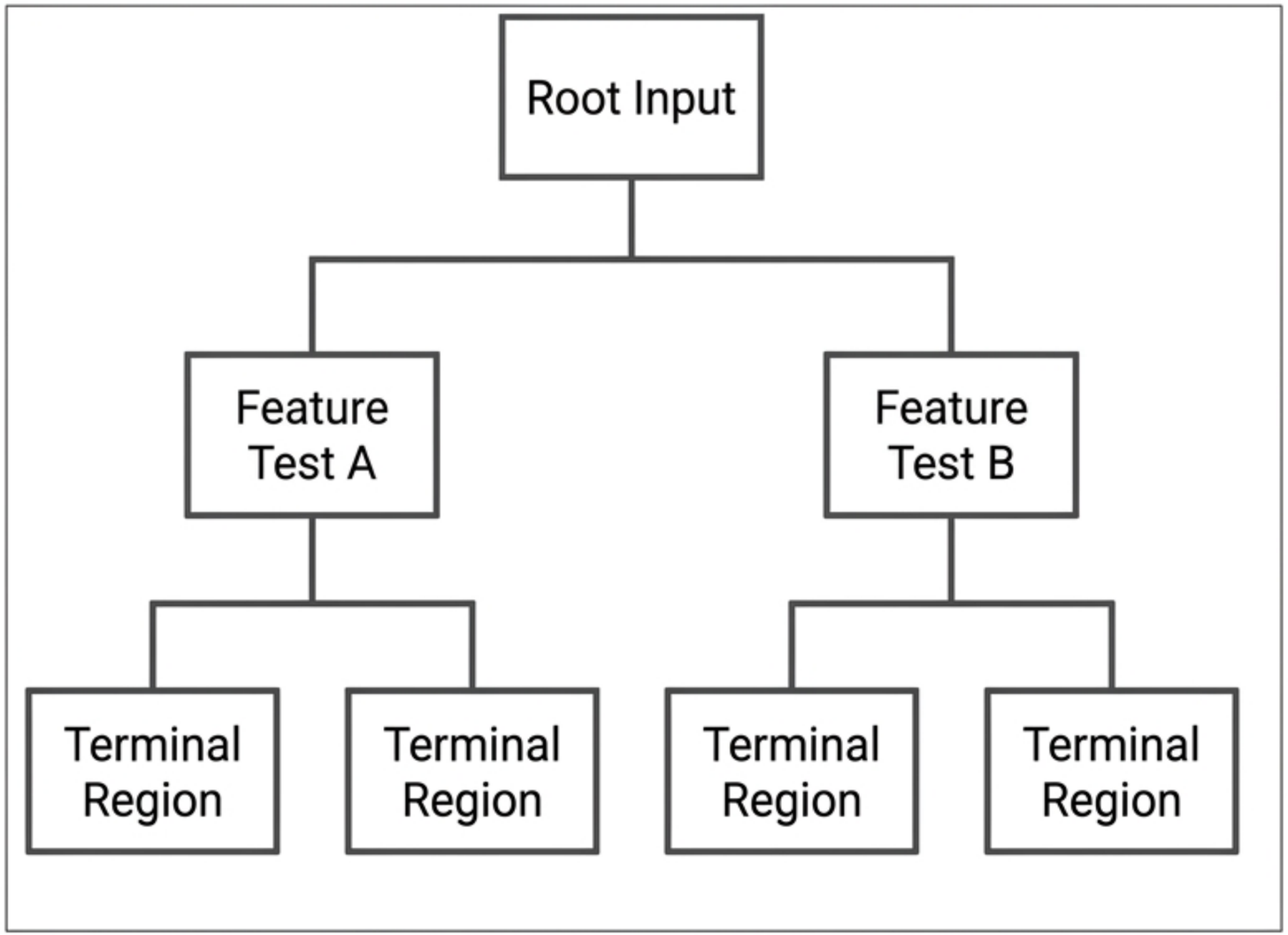
1	Decision Trees
2	Ensemble Methods
3	Application and Limitations
4	Knowledge Verification

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Section 1: Decision Trees

Purpose: Define fundamental feature-partitioning models and their predictive characteristics.

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Decision Tree Architecture

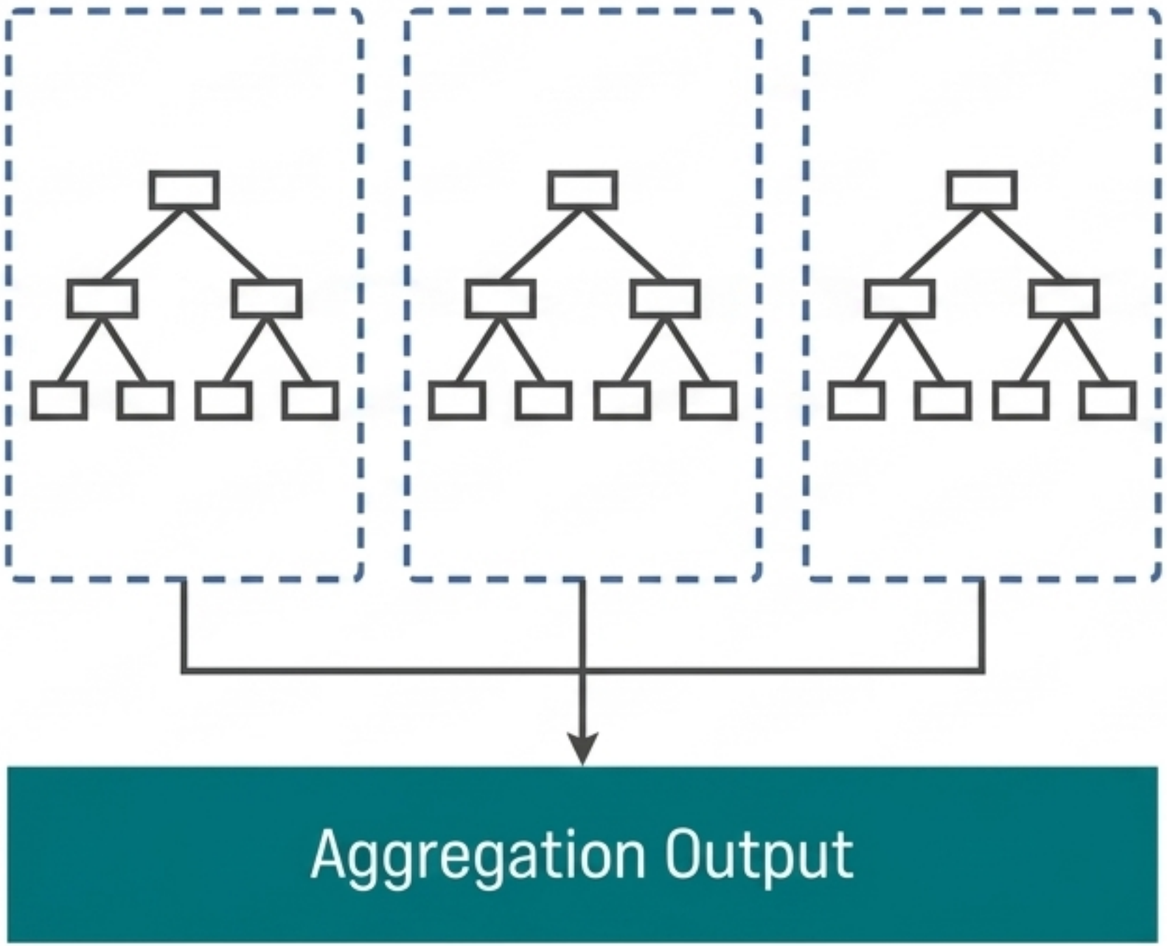
- Divide input examples into regions using feature tests.
- Generate piecewise predictions within each specific region.
- Differ fundamentally from smooth response equations.
- Small trees permit direct engineering inspection.
- Deep trees risk overfitting to training data.

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Section 2: Ensemble Methods

Purpose: Distinguish parallel randomized aggregation from sequential error reduction.

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Random Forests

- Combine multiple individual models into one ensemble.
- Utilize randomized trees to construct the forest.
- Aggregate individual predictions to form final output.
- Reduce reliance on any single decision tree.

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Sequential Boosting

- Construct models sequentially rather than in parallel.
- Target and reduce remaining objective errors progressively.
- Correct errors introduced by earlier ensemble stages.
- Differ mechanically from simple randomized aggregation.



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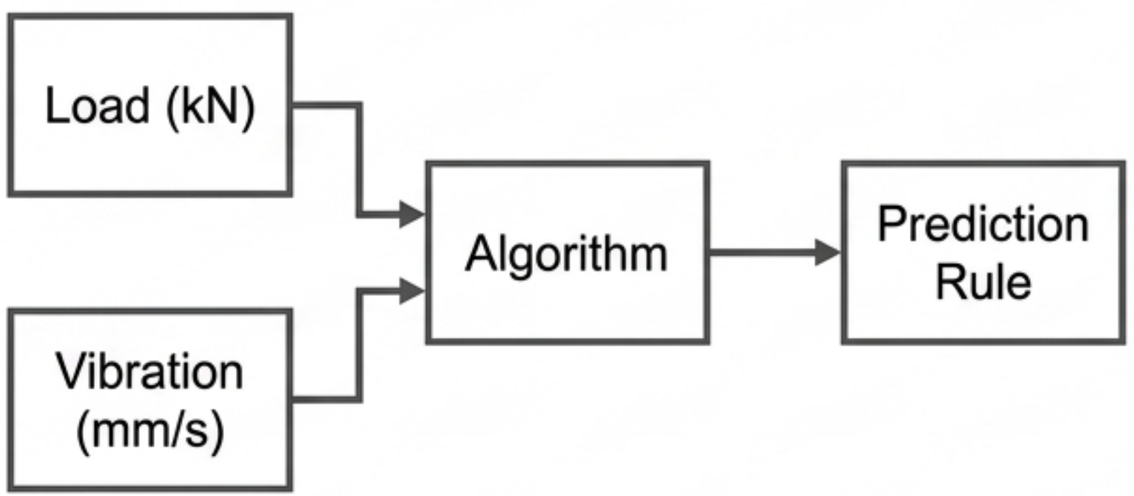
Section 3: Application and Limitations

Purpose: Establish physical boundaries and construction limitations of tree-based models.

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Prediction vs. Causation

- Algorithms learn thresholds from variables like vibration.
- Learned rules support prediction but exclude causality.
- Machine-learned thresholds do not equal manufacturer limits.
- Validate models against relevant physical operating regimes.





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Method Limitations Matrix

Method	Construction	Limitation
Small Tree	Feature partition tests	Restricted representation
Deep Tree	Many partitions	Training-specific fitting
Random Forest	Randomized aggregation	Added cost and interpretation
Boosting	Sequential additions	Validation of complexity

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Section 4: Knowledge Verification

Purpose: Assess comprehension of core tree and ensemble mechanics.

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Concept Verification

Decision trees partition inputs using defined feature tests.

Ensembles fundamentally combine multiple independent model predictions.

Boosting builds sequential models to target objective errors.

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Key Points

- Decision trees partition domains piecewise and risk overfitting.
- Forests aggregate parallel models; boosting targets sequential errors.
- Learned thresholds predict outcomes but do not prove causation.
- Core mechanics dictate strict construction and validation limits.